Light

GCSE Computer Science – Ethical, Legal and Environmental Impacts (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is 'data privacy' in the context of digital technology?

- ○ **A.** Keeping computers in a private room
- ○ **B.** The right of individuals to control how their personal information is collected, used, and shared
- ○ **C.** Making all data available to everyone
- ○ **D.** Deleting all files from a computer

Q2 (1 mark)

What is 'e-waste'?

- ○ **A.** Electronic devices that have been discarded, often containing hazardous materials requiring special disposal
- ○ **B.** Spam emails that waste time
- ○ **C.** Electricity wasted by leaving devices on standby
- ○ **D.** Digital files that take up storage space

Q3 (1 mark)

Under UK law, what is 'unauthorised access' to a computer system?

- ○ **A.** Accessing a computer system without permission, which is illegal under the Computer Misuse Act
- ○ **B.** Using your own computer at home
- ○ **C.** Forgetting your own password
- ○ **D.** Buying a second-hand computer

Q4 (1 mark)

What is an 'autonomous vehicle'?

- ○ **A.** A car that runs on electricity
- ○ **B.** Any vehicle with satellite navigation
- ○ **C.** A vehicle that can drive itself using sensors and artificial intelligence without human control

- ○ **D.** A car manufactured by robots

Q5 (1 mark)

Which of the following is an example of wearable technology?

- ○ **A.** A fitness tracker worn on the wrist that monitors heart rate and steps
- ○ **B.** A desktop computer
- ○ **C.** A television
- ○ **D.** A wireless router

Q6 (1 mark)

What is 'cloud storage'?

- ○ **A.** Storing data on servers accessed via the internet rather than on local devices
- ○ **B.** Keeping files in the sky
- ○ **C.** A type of weather prediction system
- ○ **D.** Storage that only works on rainy days

Q7 (1 mark)

Why do mobile phones raise particular privacy concerns?

- ○ **A.** Because they are small and easy to lose
- ○ **B.** Because they only work indoors
- ○ **C.** Because they cannot connect to the internet
- ○ **D.** Because they can track location, store personal data, access cameras and microphones, and are carried constantly

Q8 (1 mark)

What is a computer-based implant?

- ○ **A.** An electronic device surgically placed inside the human body to monitor health or restore function
- ○ **B.** A computer placed inside a building
- ○ **C.** Software installed on a computer
- ○ **D.** A computer keyboard

Q9 (1 mark)

What privacy risk is associated with public WiFi networks?

- ○ **A.** The WiFi signal might be too strong
- ○ **B.** Public WiFi makes phones heavier
- ○ **C.** Other users on the same network may be able to intercept unencrypted data you send and receive

- ○ **D.** WiFi networks cause phones to overheat

Q10 (1 mark)

Why do data centres have significant environmental impact?

- ○ **A.** Because they are painted with toxic colours
- ○ **B.** Because they make too much noise
- ○ **C.** Because they take up too much office space
- ○ **D.** Because they require enormous amounts of electricity for computing and cooling, contributing to carbon emissions

Q11 (1 mark)

Why might governments argue they need access to private communications data?

- ○ **A.** To read people's holiday photos
- ○ **B.** To help protect citizens from terrorism and serious crime by monitoring potential threats
- ○ **C.** Because they are curious about what people are doing
- ○ **D.** To sell the data to advertisers

Q12 (1 mark)

What is the 'digital divide'?

- ○ **A.** The gap between people who have access to digital technology and those who do not, often based on income, location, or age
- ○ **B.** A physical gap between computers
- ○ **C.** A method of splitting computer files
- ○ **D.** The space between keys on a keyboard

Q13 (2 marks)

Autonomous vehicles must be programmed to handle emergency situations. What ethical dilemma does this create?

- ○ **A.** Whether the car should play music during emergencies
- ○ **B.** Whether to use petrol or diesel fuel
- ○ **C.** How to program decisions when all options may cause harm - such as choosing between hitting pedestrians or endangering passengers - requires making moral judgements that humans typically make instinctively
- ○ **D.** What colour the car should be

Q14 (2 marks)

A company provides free fitness trackers to employees and offers rewards for meeting activity targets. What privacy concerns does this raise?

- ○ **A.** The fitness trackers might be the wrong colour
- ○ **B.** Employees might forget to charge them
- ○ **C.** The fitness trackers might be too heavy to wear

○ **D.** Employers would have continuous access to employees' health data, location, and activity patterns - potentially using this information for employment decisions, insurance purposes, or identifying 'unproductive' behaviour

Q15 (2 marks)

A UK business stores customer data using a cloud provider whose servers are located in another country. What legal issue does this create?

○ **A.** The data might get lost in translation

○ **B.** Different countries have different data protection laws - data stored abroad may be subject to foreign government access or weaker privacy protections than UK law provides

○ **C.** Cloud storage is illegal in all countries

○ **D.** The data will automatically be deleted

Q16 (2 marks)

Why do computer-based medical implants like pacemakers raise cybersecurity concerns?

○ **A.** Because wireless connectivity that enables remote monitoring and adjustment also potentially allows malicious hackers to access the device, with life-threatening consequences

○ **B.** Because they are too expensive

○ **C.** Because they are uncomfortable to wear

○ **D.** Because batteries might run out

Q17 (2 marks)

What environmental issues are associated with manufacturing digital devices?

- ○ **A.** Factories are painted with bright colours
- ○ **B.** Factories create too much paperwork
- ○ **C.** Manufacturing requires mining rare earth minerals, uses significant energy and water, generates pollution, and often involves materials sourced from environmentally damaging or ethically problematic supply chains
- ○ **D.** Manufacturing happens too quickly

Q18 (2 marks)

Some messaging apps use end-to-end encryption that even the app provider cannot decrypt. Why is this controversial?

- ○ **A.** Because encryption makes messages too long
- ○ **B.** Because encrypted messages take longer to send
- ○ **C.** Because encryption changes the font of messages
- ○ **D.** While strong encryption protects user privacy from hackers and surveillance, it also means law enforcement cannot access criminal communications even with court orders, creating tension between privacy rights and public safety needs

Q19 (2 marks)

What ethical concerns are raised about smartphone apps designed to maximise user engagement?

- ○ **A.** The apps might use too much storage space
- ○ **B.** Apps deliberately designed with addictive features like infinite scrolling, notifications, and variable rewards may harm mental health, reduce attention spans, and exploit psychological vulnerabilities for profit
- ○ **C.** The apps might have too many colours
- ○ **D.** Apps that are engaging are always beneficial

Q20 (2 marks)

If an autonomous vehicle causes an accident, who should be legally responsible?

- ○ **A.** This creates complex legal questions: the vehicle owner, the manufacturer, the software developer, or the sensor supplier could all potentially share blame - current laws designed for human drivers do not adequately address AI decision-making
- ○ **B.** Responsibility is simple and clear in all cases
- ○ **C.** The road itself is always responsible
- ○ **D.** Nobody can ever be responsible for accidents

Q21 (2 marks)

Some security researchers find vulnerabilities in software and report them to companies. Why might this 'ethical hacking' still create legal and ethical complications?

- ○ **A.** Because researchers might use the wrong keyboard
- ○ **B.** Because ethical hacking requires expensive equipment

○ **C.** Even well-intentioned security research may technically violate computer misuse laws by accessing systems without authorisation, and decisions about disclosure timing affect whether criminals or companies learn of vulnerabilities first

○ **D.** There are no complications with ethical hacking

Q22 (2 marks)

What is 'planned obsolescence' and why is it an environmental concern?

○ **A.** When devices become old-fashioned in colour

○ **B.** When devices are planned to be very durable

○ **C.** When users forget how to use their devices

○ **D.** When manufacturers deliberately design products to become outdated or fail after a certain time, forcing consumers to buy replacements - increasing e-waste and resource consumption

Q23 (2 marks)

Why do children require special consideration in discussions about digital technology and privacy?

○ **A.** Because children prefer different colours on websites

○ **B.** Children cannot fully understand or consent to data collection, are more vulnerable to manipulation, and digital footprints created in childhood follow them into adulthood when they may wish that information did not exist

○ **C.** Because children type more slowly

○ **D.** Children do not use digital technology

Q24 (3 marks)

A city installs sensors throughout public spaces to monitor traffic flow, air quality, noise levels, and pedestrian movements. This 'smart city' data is used to improve services. Analyse the benefits and privacy concerns of this approach.

○ **A.** Smart city sensors have only benefits with no privacy implications

○ **B.** Smart city sensors only create problems and have no benefits

○ **C.** Benefits include better traffic management, environmental monitoring, and evidence-based planning; concerns include constant surveillance of public movements, potential for tracking individuals, data security risks, and the gradual normalisation of pervasive monitoring that erodes expectations of anonymity in public spaces

○ **D.** Citizens have no reason to be concerned about public space monitoring

Q25 (3 marks)

Research suggests links between heavy social media use and mental health problems, particularly in young people. Analyse who bears ethical responsibility for addressing this issue: users, parents, platforms, or government regulators.

○ **A.** Only users are responsible for their own social media use

○ **B.** Government should completely ban all social media

○ **C.** Social media has no negative effects that need addressing

○ **D.** Responsibility is shared: users should moderate usage; parents should supervise children; platforms have ethical duties not to

deliberately exploit psychological vulnerabilities; and governments may need to regulate when industry self-regulation fails to protect public health - no single party bears exclusive responsibility

Q26 (3 marks)

Advances in artificial intelligence enable automation of jobs previously thought to require human skills. Analyse the potential social and economic impacts of widespread AI automation.

- ○ **A.** AI automation will only affect factory workers
- ○ **B.** Widespread AI automation may displace workers across many sectors, increase inequality if benefits flow mainly to technology owners, create pressure for retraining and education systems, and raise fundamental questions about work's role in society - while also potentially enabling new industries and improving productivity
- ○ **C.** AI cannot perform any tasks currently done by humans
- ○ **D.** Automation has no significant social implications

Q27 (3 marks)

A government introduces mandatory biometric data collection (fingerprints, facial recognition) at borders for all travellers. Analyse the security benefits and human rights concerns this raises.

- ○ **A.** Biometric border controls have no human rights implications
- ○ **B.** Biometric data collection provides perfect security with no risks
- ○ **C.** Security benefits include accurately identifying individuals, detecting fraudulent documents, and tracking known threats; concerns include mass surveillance of innocent travellers, retention of sensitive data with breach risks, potential for misuse or function creep, and disproportionate impact on minority groups if recognition systems have bias - requiring careful balancing of security and rights

○ **D.** Border security is unimportant and should not use any technology

Q28 (3 marks)

Smart home devices like voice assistants, smart TVs, and connected appliances collect data about household activities. Analyse the privacy implications of having always-listening and always-watching devices in private homes.

○ **A.** Smart home devices collect no data about users

○ **B.** Privacy concerns about smart homes are exaggerated and baseless

○ **C.** Smart home devices only activate when explicitly turned on

○ **D.** Always-on devices create continuous surveillance within traditionally private spaces, capturing intimate conversations, daily routines, and behavioural patterns; this data may be processed by companies, shared with third parties, accessed by hackers, or requested by governments - transforming the home from a private sanctuary into a monitored environment

Q29 (3 marks)

Some cryptocurrencies like Bitcoin require enormous amounts of electricity for 'mining' operations. Analyse the environmental implications of this energy consumption.

○ **A.** Cryptocurrency mining uses no significant energy

○ **B.** Cryptocurrency mining actually generates electricity

○ **C.** Bitcoin mining consumes more electricity than some countries, often from fossil fuel sources; this contributes significantly to carbon emissions for a financial system serving relatively few users - raising

questions about whether the benefits justify the environmental costs compared to more efficient alternatives

○ **D.** Environmental impact is irrelevant when evaluating technologies

Q30 (3 marks)

Social media platforms have been used to spread misinformation and influence elections. Analyse the implications of digital technology for democratic processes.

○ **A.** Digital technology has no effect on democratic processes

○ **B.** Elections should not use any technology

○ **C.** Social media only has positive effects on democracy

○ **D.** Digital platforms enable rapid information spread but also viral misinformation; targeted advertising allows manipulation of specific voter groups; filter bubbles and algorithmic content selection can polarise societies; foreign actors can interfere in elections at low cost - while platforms also enable citizen engagement and organisation; democracy faces both new opportunities and threats from digital technology

Total: **55** marks • Time: **30 min**

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